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探索性烹饪方法报告 • Exploratory Culinary Methods Report

小苏打浓度与煮制时间对意大利面拉面化效果的影响：一项 3 × 3 探索性厨房实验

Effects of Baking Soda Concentration and Cooking Time on the Ramen-Style Transformation of Spaghetti: A 3 × 3 Exploratory Kitchen Trial

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图 1 / Figure 1.
成品示例：小苏打处理后的意大利面用于汤面或拌面式呈现。图中商品包装仅用于说明实验材料；本研究与制造商无赞助或商业关系。 / Finished serving example using baking-soda-treated spaghetti. Product packaging identifies the material only; no sponsorship or commercial relationship is claimed.

摘要与关键词 / Abstract and Keywords

摘要： 在难以获得新鲜碱水面或拉面的环境中，商品干制意大利面常被用作替代材料。本探索性厨房实验采用 3 × 3 设计，比较每升水中加入 2、4、6 g 小苏打，并分别煮制 10、12、15 分钟的九种组合。作者依据弹性、顺滑度、碱香纯度和接近拉面的程度进行结构化感官评价，并给出 10 分制综合评分。4 g/L、15 分钟处理得分最高（9.5/10），被描述为颜色金黄、Q 弹顺滑、碱香适中且无明显苦味；4 g/L、12 分钟得分 8.5/10，更适合偏硬口感。2 g/L 处理整体变化有限；6 g/L 随时间延长出现苦味、涩感、黏口和肥皂味。结果表明，在本次单人、探索性厨房条件下，4 g/L 小苏打与 15 分钟煮制是最优组合。

关键词： 碱水面；小苏打；意大利面；拉面替代；面条质构；厨房实验

Abstract: Commercial dried spaghetti can serve as a substitute when fresh alkaline noodles are difficult to obtain. This exploratory kitchen trial used a 3 × 3 design combining baking soda concentrations of 2, 4, and 6 g/L with cooking times of 10, 12, and 15 minutes. Treatments were assessed for elasticity, smoothness, alkaline aroma, and perceived similarity to ramen-style noodles, followed by an overall score on a 10-point scale. The highest score, 9.5/10, was assigned to 4 g/L for 15 minutes, described as golden, springy, smooth, appropriately alkaline, and not noticeably bitter. The 4 g/L, 12-minute treatment scored 8.5/10 and retained a firmer center. Treatments at 2 g/L produced limited transformation, whereas prolonged cooking at 6 g/L produced bitterness, astringency, stickiness, and a soap-like odor. Within the limits of a single-observer exploratory trial, 4 g/L for 15 minutes gave the most favorable result.

Keywords: alkaline noodles; baking soda; spaghetti; ramen substitute; noodle texture; culinary methods

1. 背景 / Background

海外生活或食材获取受限时，方便获得的干制意大利面可作为亚洲面条的替代基础。小苏打（碳酸氢钠）使煮面水呈碱性，并可能改变面条颜色、气味和口感。本实验旨在以结构化厨房测试比较小苏打浓度与煮制时间的组合，寻找兼顾弹性、顺滑度与适度碱香的实用条件。

When fresh Asian alkaline noodles are inaccessible, widely available dried spaghetti offers a practical starting material. Adding baking soda makes the cooking water alkaline and may alter color, aroma, and texture. The objective was to compare concentration-time combinations and identify a practical condition balancing springiness, smoothness, and moderate alkaline character.

2. 材料与方法 / Materials and Methods

材料 / Materials. 商品干制硬质小麦意大利面、小苏打和饮用水。每组使用 1 L 水；原帖未记录每组干面质量。商品由作者自行购买。 / Commercial dried durum-wheat spaghetti, baking soda, and water were used. Each treatment used 1 L water; the dry spaghetti mass per treatment was not documented. The product was independently purchased.



图 2 / Figure 2. 实验材料与 4 g 小苏打称量。 / Commercial dried spaghetti and measurement of 4 g baking soda.

实验设计 / Design. 小苏打浓度为 2、4、6 g/L；煮制时间为 10、12、15 分钟，共九种组合。水量统一为 1 L，中火煮制。原帖提及普通意大利面作为概念性对照，但未提供独立对照评分。 / Baking soda concentrations were 2, 4, and 6 g/L, combined with cooking times of 10, 12, and 15 minutes. Each treatment used 1 L water over medium heat. Ordinary spaghetti was mentioned as a conceptual control, but no separate control score was supplied.

评价 / Evaluation. 作者从弹性、顺滑度、碱味纯度和接近拉面的程度四方面进行主观评价，并给出 10 分制综合评分。未进行盲评、重复试验或仪器测量。 / The author evaluated elasticity, smoothness, alkaline-aroma quality, and ramen-like similarity, then assigned an overall 10-point score. No blinding, replication, or instrumental testing was reported.

3. 结果 / Results

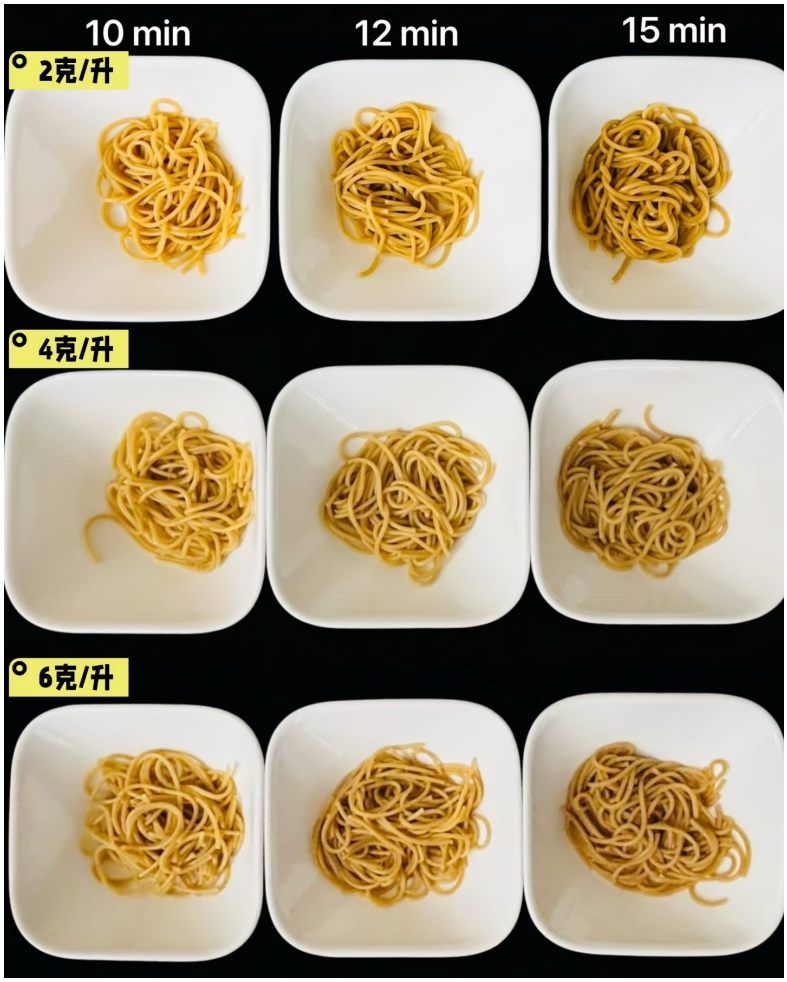


图 3 / Figure 3. 九种小苏打浓度与煮制时间组合的外观对比。图像显示碱浓度和煮制时间增加时颜色总体加深，但照片未经过颜色仪校准。 / Visual comparison of nine treatments. Color generally darkened with concentration and time, but images were not instrumentally color-calibrated.

表 1 / Table 1. 九种处理的综合感官评分（10 分制） / Overall descriptive scores for the nine treatments (10-point scale).

小苏打 Baking soda	10 min	12 min	15 min
2 g/L	5.0	6.0	6.5
4 g/L	7.0	8.5	9.5
6 g/L	6.0	5.0	3.5

详细观察 / Detailed Observations

处理 / Treatment	主要观察 / Main observation
2 g/L • 10 min	5.0：仍像意面，有硬芯，几乎无碱香。 / Still spaghetti-like; firm core; almost no alkaline aroma.
2 g/L • 12 min	6.0：弹性一般，碱香很淡，拉面感弱。 / Average elasticity; faint alkaline aroma; weak ramen character.
2 g/L • 15 min	6.5：熟透但偏软，筋道不足。 / Fully cooked but soft and insufficiently springy.
4 g/L • 10 min	7.0：外弹内硬，中心仍有白芯。 / Elastic exterior but firm, pale center.
4 g/L • 12 min	8.5：接近拉面，很弹，芯略硬。 / Close to ramen; very elastic; slightly firm center.
4 g/L • 15 min	9.5：最像拉面，Q 弹顺滑，碱香适中。 / Most ramen-like; springy, smooth, balanced alkaline aroma.
6 g/L • 10 min	6.0：碱味偏重、芯硬、回口微苦。 / Strong alkalinity; firm center; mildly bitter finish.
6 g/L • 12 min	5.0：苦涩，仍有弹性但黏口。 / Bitter and astringent; elastic but sticky.
6 g/L • 15 min	3.5：肥皂味、表面发黏、苦味明显。 / Soap-like odor, sticky surface, pronounced bitterness.



图 4 / Figure 4. 原始评分矩阵和中文描述。 / Original score matrix and Chinese treatment notes.

4. 实用方法 / Practical Procedure

步骤 / Step	操作 / Procedure
1	向 1 L 水中加入 4 g 小苏打并加热至沸腾。 / Add 4 g baking soda to 1 L water and bring to a boil.
2	水开后加入意大利面。 / Add spaghetti after the water boils.
3	煮满 15 分钟；偏好更硬口感者可尝试 12 分钟。 / Cook for 15 minutes; 12 minutes may suit a firmer preference.
4	使用容量较大的锅并留意泡沫。 / Use a larger pot and monitor foaming.
5	捞出后用于汤面或拌面；根据菜式调整调味。 / Drain and use in soup or mixed-noodle dishes; season to suit the preparation.



图 5 / Figure 5. 碱性煮面水出现泡沫和黄褐色。该图仅记录外观；实验未测量 pH、色差或煮制损失。 / Alkaline cooking water developed foam and a yellow-brown color. Appearance only; pH, color difference, and cooking loss were not measured.

5. 讨论 / Discussion

评分呈现明显的浓度与时间组合效应：2 g/L 随时间增加仅小幅改善；4 g/L 随时间增加持续改善；6 g/L 则随时间增加而下降。最优点出现在中等浓度与最长煮制时间，而高浓度长时间处理产生明显不良气味和口感。该模式说明“更多碱”并不等同于“更像拉面”，碱浓度与面芯熟化需要共同平衡。

The score pattern suggests a concentration-time interaction. Scores rose modestly over time at 2 g/L, rose strongly at 4 g/L, and declined over time at 6 g/L. The best outcome occurred at the middle concentration and longest cooking time, whereas prolonged high-alkali treatment produced undesirable flavor and surface

properties. More alkali did not necessarily create a better ramen-like result; concentration had to be balanced with center softening.

6. 局限性 / Limitations

本研究属于单人探索性厨房实验。原始材料未报告独立重复次数、每组干面质量、盲评流程或独立感官评审人数。评分为作者的结构化主观判断，未进行统计检验。实验也未测量 pH、仪器质构、含水率、煮制损失或仪器色差。照片可能受到光线、曝光、相机处理和拍摄时间影响。对照组虽在原帖中提及，但没有单独评分。因此，本结果应理解为可复现性有待进一步验证的实用厨房观察，而非普遍性结论。

This was a single-observer exploratory kitchen trial. Independent replication, dry pasta mass per treatment, blinding, and an external sensory panel were not documented. Scores were structured subjective judgments without statistical testing. pH, instrumental texture, moisture, cooking loss, and calibrated color were not measured. Photography may have been affected by lighting and camera processing. A control was mentioned but not independently scored. The findings are practical observations that require replication rather than universal conclusions.

7. 结论 / Conclusion

在本次 3 × 3 厨房实验中，4 g/L 小苏打、煮制 15 分钟获得最高综合评分 9.5/10；4 g/L、12 分钟则提供更偏硬的选择。2 g/L 改变有限，6 g/L 尤其在长时间煮制后产生苦涩、黏口和肥皂味。对希望利用商品意大利面制作拉面风格替代品的家庭烹饪者而言，4 g/L、15 分钟可作为透明且易操作的起始方法。

In this 3 × 3 kitchen trial, 4 g/L baking soda for 15 minutes received the highest score, 9.5/10, while 4 g/L for 12 minutes offered a firmer alternative. The 2 g/L treatments produced limited transformation, and 6 g/L, particularly with prolonged cooking, produced bitterness, stickiness, and a soap-like odor. For home cooks seeking a ramen-style substitute using commercial spaghetti, 4 g/L for 15 minutes is a practical starting method.

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